



BAULKHAM HILLS HIGH SCHOOL

Half -Yearly 2013
YEAR 11 TASK 1

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 1 hour and 30 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in Questions 6-11
- Marks may be deducted for careless or badly arranged work

Total marks – 51

Exam consists of 5 pages.

This paper consists of TWO sections.

Section 1 – Page 2 (5 marks)

- Attempt Question 1-5

Section II – Pages 3-5 (46 marks)

- Attempt questions 6-11

Answer all questions in the appropriate space in the Answer booklet provided.

Section I – Multiple Choice - 5 marks

1. $a + b\sqrt{2} = \frac{2}{3+\sqrt{2}}$ find a and b .

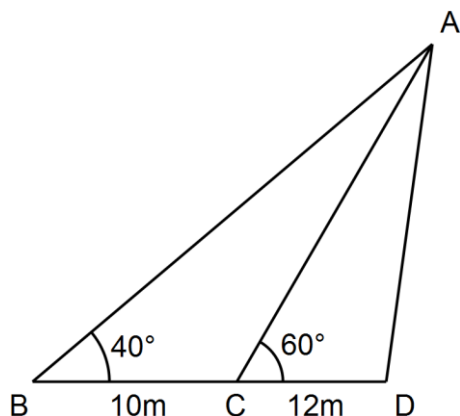
(A) $a = \frac{6}{7}, b = -\frac{2}{7}$

(B) $a = \frac{6}{7}, b = \frac{2}{7}$

(C) $a = \frac{6}{5}, b = \frac{2}{5}$

(D) $a = \frac{6}{5}, b = -\frac{2}{5}$

2.



In the diagram $AC =$

(A) $12 \times \frac{\sin 20^\circ}{\sin 40^\circ}$

(B) $10 \times \frac{\sin 40^\circ}{\sin 20^\circ}$

(C) $12 \times \frac{\sin 40^\circ}{\sin 20^\circ}$

(D) $10 \times \frac{\sin 20^\circ}{\sin 40^\circ}$

3. If P divides AB externally in the ratio $5:2$, then B divides PA in the ratio

(A) $3:2$

(B) $3:-2$

(C) $2:3$

(D) $2:-3$

4. Determine the number of solutions for $\cos x - 4 \sec x = 0$ for $-180^\circ \leq x \leq 180^\circ$.

(A) 0

(B) 1

(C) 2

(D) 3

5. There are p triangles that can be formed from the 8 vertices of a given cube, and q of these triangles are equilateral. The values of p and q are:

(A) $p = 56, q = 6$

(B) $p = 20, q = 6$

(C) $p = 56, q = 8$

(D) $p = 20, q = 8$

End of Section 1

Section II – Extended Response

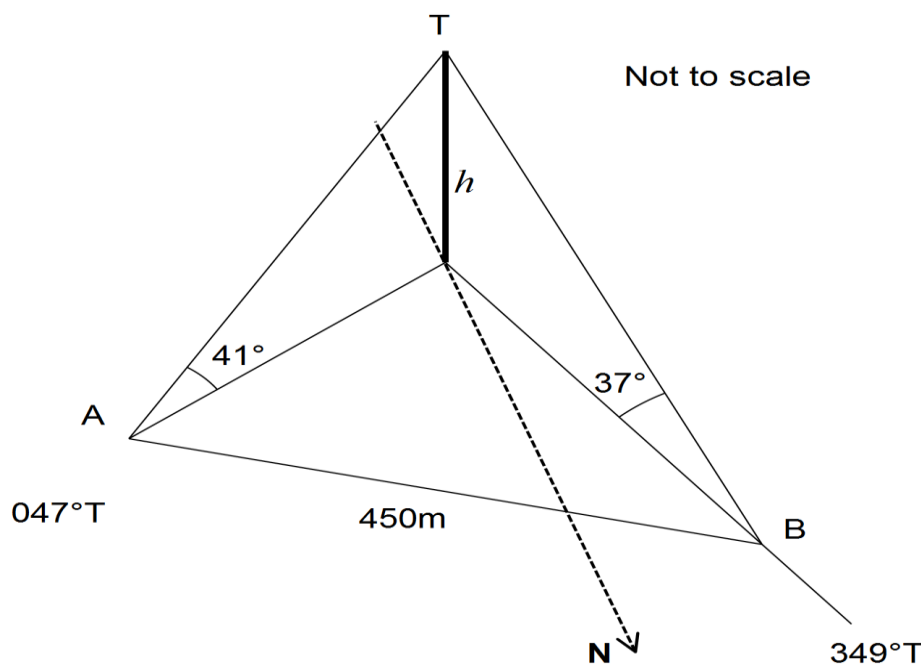
Attempt questions 6-11.

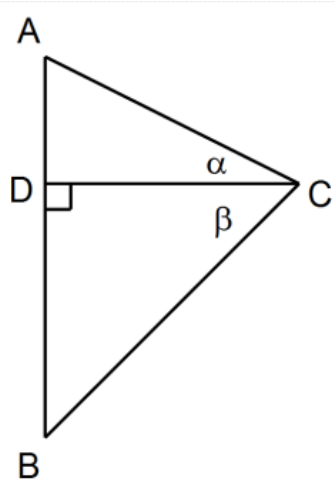
All necessary working should be shown in every question.

Question 6 (8 marks)		Marks
a)	Solve $\frac{4}{x-1} \geq 1$	3
b)	Simplify $\frac{a^n + a^{n-2}}{a^{n-1}}$	2
c)	Solve $2 \cos^2 x - \sin x - 1 = 0$ for $0^\circ \leq x \leq 360^\circ$	3

Question 7 (9 marks)		
a)	The point $(6, k)$ is 8 units from the straight line, $3x + 4y + 2 = 0$, find k .	3
b)	Given the points $A(-5, 1)$ and $B(11, 9)$, find the point P which divides the interval AB externally in the ratio 5:3	3
c)	(i) Sketch $y = 2x - 4 $	1
	(ii) Find the values for c for which $ 2x - 4 = x + c$ has two solutions.	2

Question 8 (8 marks)		
a)	A committee of 6 people is to be selected from 6 men and 4 women	
	(i) How many different committees can be formed?	1
	(ii) How many committees are possible if the men outnumber the women?	2
b)	If $4x^2 + 4xy + y^2 = 0$ find the value of $\frac{x-y}{x+y}$ where $x \neq y$	2
c)	Prove that $\frac{1 + \cos \theta}{1 - \sin \theta} - \frac{1 - \cos \theta}{1 + \sin \theta} = 2 \sec \theta (1 + \tan \theta)$	3

Question 9 (7 marks)		Marks
a)	From the top of a tower T, two markers A and B can be seen on horizontal ground.  Marker A lies on a bearing of $047^\circ T$ from the tower. Marker B has a bearing of $349^\circ T$ from the tower. The angles of elevation from A and B to the top of the tower are 41° and 37° respectively. If AB is 450m, find h, the height of the tower, to the nearest metre.	3
b)	(i) How many different arrangements of the letters of the word PARALLEL are there? (ii) How many of these arrangements begin and end with the letter L?	2 2

Question 10 (6 marks)		
a)	 Given triangle ABC where D is a point on AB such that $CD \perp AB$, $\angle ACD = \alpha$ and $\angle DCB = \beta$. Prove that $DB = \frac{AD \cos \alpha \sin \beta}{\cos \beta \sin \alpha}$	3
b)	Given x and y are rational, solve the following for x and y. $xy + \sqrt{9x^2 + y^2} = 8 + 2\sqrt{13}$	3

Question 11 (8 marks)		Marks
a)	(i) Graph the function $y = 2 x - 1 + x - 4 $	3
	(ii) Use your graph to solve $2 x - 1 + x - 4 \leq 6$	1
b)	Three identical cubes of side 1cm are placed together as shown in the diagram. (i) Find the exact length of AB (ii) Find $\angle ABC$ to the nearest degree.	2 2

End of Exam

	BANK	
9	Solve $\frac{12}{x^2-x} \leq 1$	4
3	If $\frac{4}{x-1} \geq 1$ then (A) $x < 1$ or $x \geq 5$ (B) $1 \leq x \leq 5$ (C) $1 < x \leq 5$ (D) $x \leq 1$ or $x \geq 5$	
7b	By graphing the function $y = x - 2 + x - 4 $ or otherwise solve $ x - 2 + x - 4 = 6$	3
7a	Find the distance between the two straight lines $3x - 5y + 11 = 0$ and $5y - 3x + 6 = 0$	3
	A group of 12 people is to be seated at a long table with 6 seats each side. There are 4 people who wish to be on one side of the table and 3 people who wish to be on the other side. How many seating arrangements are there?	3
9b	The letters A A A B B C C D E E F F F are arranged in all possible ways. If one of the arrangements is chosen at random what is the probability that it starts with ABCD	3
	How many sets of 5 bands can be formed from 5 lead guitars, 5 bass guitars, 5 drummers and 5 pianists, if each band has to have at least one player of each instrument.	3
	If $\frac{12}{x^2-x} \geq 1$ then (A) $-3 \leq x \leq 0$ or $1 \leq x \leq 4$ (B) $-3 \leq x \leq 0$ (C) $-3 \leq x < 0$ or $1 < x \leq 4$ (D) $x \leq -3, 0 < x < 1$ or $x \geq 4$	

①

$$a + b\sqrt{2} = \frac{2}{3+\sqrt{2}} \times \frac{3-\sqrt{2}}{3-\sqrt{2}}$$

1 EACH

$$= \frac{6 - 2\sqrt{2}}{9-2}$$

$$= \frac{6}{7} - \frac{2\sqrt{2}}{7}$$

$$a = \frac{6}{7} \quad b = -\frac{2}{7}$$

— (A)

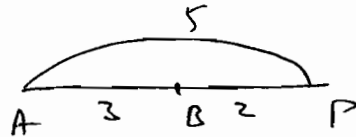
②

$$\frac{AC}{n40} = \frac{10}{n20}$$

$$AC = \frac{10n40}{n20}$$

— (B)

③



$$PB : BA = 2 : 3$$

— (C)

④

$$\cos x - 4\sec x = 0$$

(A)

$$\cos x = \frac{4}{\cos x}$$

$$\cos^2 x = 4$$

no soln

⑤

$$P = 8C_2 = 56.$$

(C)

$$\text{each day} \rightarrow 2 \text{ } \Delta \text{ } \Delta$$

$$\therefore \text{each fence} \rightarrow 2 \times 2 = 4 \text{ } \Delta \text{ } \Delta$$

$$6 \text{ fences} \times 4 = 24 \quad \div 3 (\text{repetition})$$

$$9 = 8$$

MC 1 - A
2 - B
3 - C

4 - A
5 - C

6a) $\frac{4}{x-1} \geq 1 \quad x \neq 1 \quad \perp$

$$4(x-1) \geq (x-1)^2$$

$$(x-1)^2 - 4(x-1) \leq 0 \quad \perp$$

$$(x-1)(x-5) \leq 0 \quad \perp$$

$$1 < x \leq 5$$



6b) $\frac{a^n + a^{n-2}}{a^{n-1}} \quad \perp$

$$= \frac{a^2(a^2+1)}{a^{n-1}}$$

$$= \frac{a^2+1}{a} \quad \text{or} \quad a + \frac{1}{a} \quad \perp$$

6c) $2\cos^2 x - \sin x - 1 = 0$

$$2(1 - \sin^2 x) - \sin x - 1 = 0 \quad \perp$$

$$2\sin^2 x + \sin x - 1 = 0$$

$$(2\sin x - 1)(\sin x + 1) = 0$$

$$\sin x = \frac{1}{2} \quad \text{or} \quad -1 \quad \perp$$

$$\therefore x = 30^\circ, 150^\circ \text{ or } 270^\circ \quad \perp$$

$$\begin{aligned}
 \text{a)} \quad D &= \left| \frac{Ax_1 + By_1 + C}{\sqrt{A^2 + B^2}} \right| \\
 &= \left| \frac{3 \times 6 + 4 \times k + 2}{\sqrt{3^2 + 4^2}} \right| = \frac{20 + 4k}{5} \\
 &= \left| \frac{20 + 4k}{5} \right| \quad \underline{1}
 \end{aligned}$$

$$\therefore \left| \frac{20 + 4k}{5} \right| = 8$$

$$\therefore |20 + 4k| = 40$$

$$(5 + k) = \pm 10 \quad \underline{1} \quad \underline{1}$$

$$k = 5 \text{ or } -15.$$

$$k = 5 \text{ or } -15$$

2 marks.

$$\text{b)} \quad A(-5, 1) \quad B(11, 9) \quad m:n = 5:-3$$

$$x = \frac{mx_1 + nx_2}{m+n}$$

$$y = \frac{my_1 + ny_2}{m+n}$$

$$= \frac{-3 \times -5 + 5 \times 11}{5-3}$$

$$= \frac{-3 \times 1 + 5 \times 9}{5-3} \quad \underline{1}$$

$$= \frac{70}{2}$$

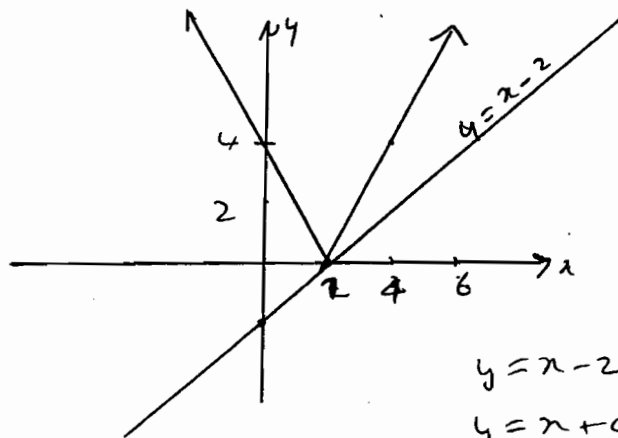
$$= \frac{42}{2}$$

$$= 35 \quad \underline{1}$$

$$= 21 \quad \underline{1}$$

$$\therefore P \text{ is point } (35, 21)$$

c)



$$y = x - 2 \rightarrow \text{one solution} \quad \underline{1}$$

$$y = x + c \quad c > -2 \text{ has 2 solutions}$$

2

$$84) \quad 1) \quad {}^{10}C_6 = 210 \quad \underline{1}$$

$$11) \quad 4m2w + 5m1w + 6m$$

$$= {}^6C_4 \cdot {}^4C_2 + {}^6C_5 \cdot {}^4C_1 + {}^6C_6 \quad \underline{1}$$

$$= 15 \times 6 + 6 \times 4 + 1$$

$$= 90 + 24 + 1$$

$$= 115 \quad \underline{1}$$

$$b) \quad 4x^2 + 4xy + y^2 = 0$$

$$2x + y = 0$$

$$y = -2x \quad \underline{1}$$

$$\therefore \frac{x-y}{x+y} = \frac{x-(-2x)}{x-2x}$$

$$= \frac{3x}{-x}$$

$$= -3 \quad \underline{1}$$

$$c) \text{ LHS} = \frac{1+\cos\theta}{1-\sin\theta} - \frac{1-\cos\theta}{1+\sin\theta}$$

$$= \frac{(1+\cos\theta)(1+\sin\theta) - (1-\cos\theta)(1-\sin\theta)}{1-\sin^2\theta} \quad \underline{1}$$

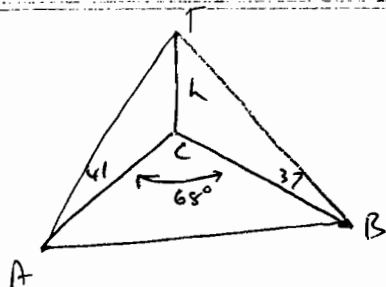
$$= \frac{\cancel{1}\cos\theta + \cancel{\sin\theta}\cancel{\sin\theta} + \cancel{1}\sin\theta + \cancel{\cos\theta}\cancel{\sin\theta} - \cancel{1} + \cancel{\cos\theta} + \cancel{\sin\theta}\cancel{\sin\theta} - \cancel{1}\cos\theta}{\cos^2\theta}$$

$$= \frac{2}{\cos^2\theta} (\cos\theta + \sin\theta)$$

$$= \frac{2}{\cos^2\theta} (1 + \tan\theta) \quad \underline{1}$$

$$= \text{RHS}$$

multiply



$$\tan 41 = \frac{h}{AC}$$

$$\tan 37 = \frac{h}{BC}$$

$$\therefore AC = h \tan 49^\circ$$

$$BC = h \tan 53^\circ \quad \underline{1}$$

$$\angle ACB = 47 + 11 = 58^\circ$$

$$AB^2 = AC^2 + BC^2 - 2 \cdot AC \cdot BC \cdot \cos \angle ACB$$

$$\therefore 450^2 = h^2 \tan^2 49^\circ + h^2 \tan^2 53^\circ - 2 \cdot h^2 \tan 49^\circ \tan 53^\circ \cos 58^\circ \quad \underline{1}$$

$$= h^2 (\tan^2 49^\circ + \tan^2 53^\circ - 2 \tan 49^\circ \tan 53^\circ \cos 58^\circ)$$

$$\therefore h = \frac{450}{\sqrt{\tan^2 49^\circ + \tan^2 53^\circ - 2 \tan 49^\circ \tan 53^\circ \cos 58^\circ}}$$

$$= 372 \text{ m.} \quad \underline{1}$$

b1)
$$\begin{matrix} P & A & R & C & E \\ & A & & L & \\ & & & L & \end{matrix}$$

$$N_0 = \frac{8!}{2! \cdot 3!} \quad \underline{1} = 3360 \quad \underline{1}$$

ii) $L - - - - - L \rightarrow \frac{6!}{2!} \quad \underline{1} = 360 \quad \underline{1}$

(10) a)

$$\text{in } \triangle ADC \quad \tan \alpha = \frac{AD}{DC}$$

$$\begin{aligned}\therefore DC &= \frac{AD}{\tan \alpha} \\ &= AD \frac{\cos \alpha}{\sin \alpha} \quad \perp\end{aligned}$$

$$\text{in } \triangle BDC \quad \tan \beta = \frac{DB}{DC}$$

$$\begin{aligned}\therefore DC &= \frac{DB}{\tan \beta} \\ &= \frac{DB \cos \beta}{\sin \beta} \quad \perp\end{aligned}$$

$$\therefore \frac{DB \cos \beta}{\sin \beta} = \frac{AD \cos \alpha}{\sin \alpha} \quad \perp$$

$$DB = \frac{AD \cos \alpha \sin \beta}{\cos \beta \sin \alpha}$$

b) $xy = 8 \quad \text{--- (1)}$

$9x^2 + y^2 = 52 \quad \text{--- (2)} \quad \perp$

Subst $y = \frac{8}{x} \rightarrow \text{(2)}$

$$\therefore 9x^2 + \frac{64}{x^2} = 52$$

$$\therefore 9x^4 - 52x^2 + 64 = 0$$

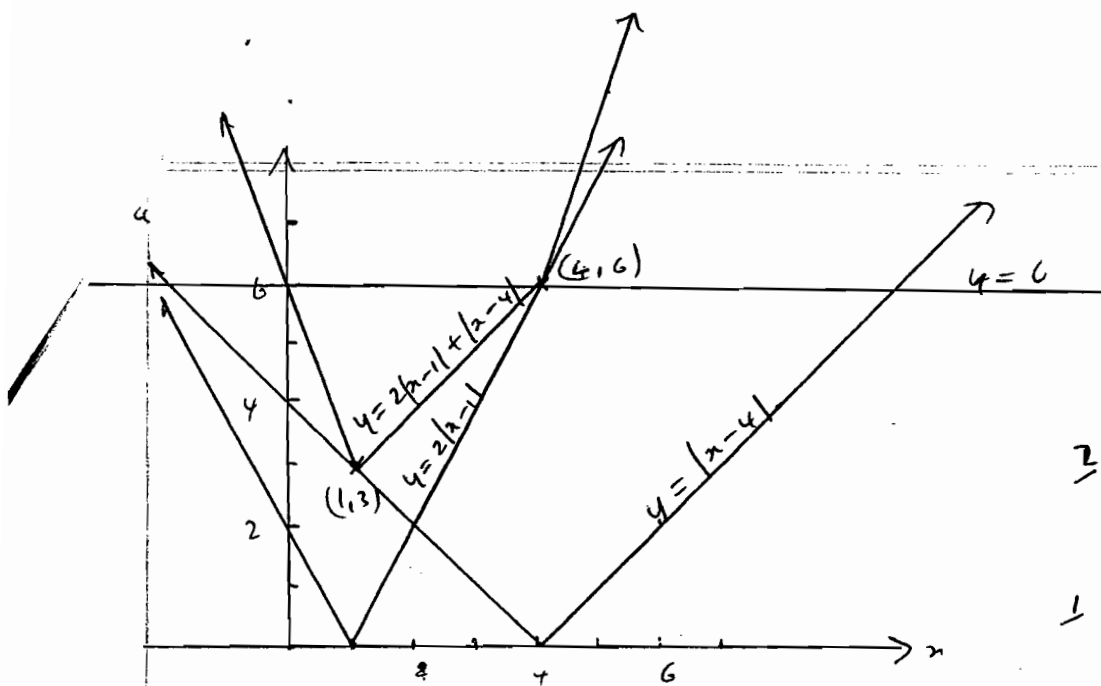
$$(9x^2 - 16)(x^2 - 4) = 0 \quad (\text{or let } y = x^2 \text{ etc})$$

$$\therefore x^2 = \frac{16}{9} \text{ or } 4 \quad \perp$$

$$x = \frac{4}{3} \text{ or } 2$$

$$\therefore y = \frac{16}{3} \text{ or } 4 \text{ from (1)}$$

$$\therefore \text{solution } \left(\frac{4}{3}, \frac{16}{3}\right) \text{ or } (2, 4) \quad \perp$$



2 CORRECT SHMDF

1 SHOWING (0, 6) (1, 3) (4, 6)

11) $0 \leq x \leq 4$

1

11 b

$$AB^2 = 2^2 + 2^2 + 1^2 = 9$$

1

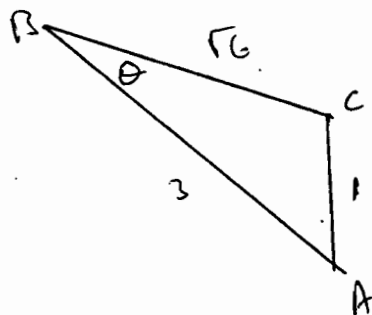
$$\therefore AB = 3$$

1

$$BC^2 = 2^2 + 1^2 + 1^2 = 6$$

$$\therefore BC = \sqrt{6}$$

1



$$\cos \angle ABC = \frac{(\sqrt{6})^2 + 3^2 - 1}{2 \cdot 3 \cdot \sqrt{6}}$$

$$= \frac{14}{6\sqrt{6}}$$

$$\therefore \angle ABC = 17.7^\circ$$

1

$= 18^\circ$ to nearest degree